Audio Deepfake Detection: A Systematic Literature Review of Acoustic, Deep Learning, Self-Supervised, and Audio-LLM-Based Approaches

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Abstract

Deepfake technology is now a worry and the misuse of synthetic voice generation brings serious dangers such as fraud, identity impersonation, financial crime, misinformation and political manipulation. I see that Deepfake technology keeps growing and voice generation is a new threat. This systematic literature review looks at how audio deepfake detection has changed. It starts with handcrafted features and machine learning then goes to deep learning Transformer-based methods, self-supervised learning and the newest Large Language Model and Audio-LLM techniques. We went through 82 studies that were published from 2023 to 2026 in places such as Elsevier, Wiley, IEEE, ACM, ACL, MDPI, Springer Nature and ICLR. The studies were examined using datasets, features, methodologies, evaluation settings, results, limitations and research contributions. The review shows a move from handcrafted acoustic features toward pretrained speech representations and more recently, toward semantic and Audio-LLM-based approaches. Out of the 82 studies 15 look at Large Language Model, Audio-LLM or multimodal language-model approaches. The findings say that mixing evidence, self-supervised speech representations and Audio-LLM reasoning is a hopeful path, for building sturdy, generalizable and explainable AI-generated voice detection systems.

1 Introduction

Recent progress in intelligence now allows creators to produce very realistic human speech and other audio types. Modern text-to-speech systems, voice-conversion tools, neural audio codecs and generative models can make speech that sounds like real recordings. These tools help people communicate, improve accessibility entertain and create content. They also introduce security and forensic dangers. Synthetic voices can be used for impersonation, fraud, social engineering, misinformation and other harmful actions. As speech-generation systems become common reliable detection of audio deepfakes becomes a growing research topic.

Audio deepfake detection is about finding out if a speech recording is real or made by a computer or changed in some way. In the past people used special sound features that they created themselves and basic machine learning or deep networks. More recent research has started to use models that work with patterns, systems that work directly with sound waves models that remember information over time models that use attention like transformers and systems that learn without needing labels. For example Li and team looked at deepfake .detection in voices using HuBERT showing how computer generated sound features can help with this job. Almutairi and Elgibreen also looked at computer generated learning to find Arabic speech showing how important it is to use these methods for languages other, than English. Other research has looked at patterns, sound features, voice samples and combining different methods to find better clues[34, 5, 18, 9, 46, 4].

a growing gap between how deepfakesre made and how they are detected. A detector that works well on a group of generators or datasets might do fine in controlled settings but it can fail when it faces new synthesis systems, different languages, unknown speakers, various codecs, different recording conditions or real-world communication environments. Research has looked at detecting deepfakes across languages, types of deepfake generators big-scale benchmarks and how well models generalize across languages in audio deepfake detection. Recently researchers have started using foundation models and in-context learning to make audio deepfake detection systems

better at generalizing and being more interpretable[6, 7, 59]. These changes show how quickly audio deepfake detection is changing. They also call for a review of advances from 2023, to 2026.

Dataset design is another important factor in audio deepfake detection. Several recent studies have focused on constructing datasets that represent newer generation technologies and more realistic conditions. Codecfake introduced a dataset specifically targeting audio deepfakes generated using neural audio codecs [37], while its subsequent expanded study investigated countermeasures for universal deepfake audio detection [57]. SpeechFake provided a large-scale multilingual speech deepfake dataset incorporating recent generation methods [26]. SynSpeech introduced another dataset and benchmark for fake speech detection [43]. More recent datasets, including AUDETER and P2V, have further emphasized open-world detection and robustness against public-voice perturbations [51, 21]. These developments demonstrate that detector performance cannot be evaluated independently of the generation methods, languages, speakers, and acoustic conditions represented in the evaluation data.

Another important direction is the development of robust detection systems. Real-world audio can contain noise, reverberation, compression, codec artifacts, partial manipulation, and other distortions that are not necessarily present in laboratory datasets. Researchers have therefore investigated approaches based on denoising, noise augmentation, adversarial robustness, multi-level self-supervised features, waveform representations, and time-frequency modeling [30, 44, 50, 47, 32]. Other work has investigated partial fake speech and temporal localization, moving beyond the simple binary question of whether an entire recording is fake [3, 56]. These studies indicate that practical audio deepfake detection requires robustness as well as high classification accuracy.

At the same time, the field is moving beyond conventional discriminative models toward large language models (LLMs) and multimodal or audio-language models. Audio-language models provide an opportunity to process acoustic information together with higher-level semantic or linguistic information. Gu et al. explored the use of audio large language models for audio deepfake detection, representing an important transition toward LLM-based detection [25]. Chuchra et al. investigated multimodal large language models for audio deepfake detection, further demonstrating the potential of multimodal reasoning in this domain [12]. Venaik et al. proposed an LLM-integrated framework for continual adaptation in spoofed speech detection [48], while Chou et al. investigated in-context learning with comparison guidance for audio deepfake detection [11]. Gohari et al. further explored audio-language models for interpretable detection of manipulated singing voices [23]. These studies suggest that LLM-based systems may provide capabilities beyond conventional binary classification, including contextual reasoning, comparison, interpretation, and adaptation.

Despite these advances, the literature remains fragmented across different model families, datasets, languages, evaluation settings, and objectives. Conventional deep learning approaches, self-supervised models, transformer architectures, multimodal systems, LLM-based methods, dataset-oriented studies, and robustness-oriented approaches are often investigated separately. Furthermore, the rapid appearance of new datasets and generation technologies makes it difficult to determine which findings remain reliable across different experimental conditions. Recent comparative and systematic studies have begun to address this problem. For example, Li et al. presented a systematic analysis of audio deepfake detection in relation to generative and detection models [35], while Firc et al. developed an evaluation framework for comparing state-of-the-art deepfake speech detectors [19]. The Speech DF Arena further reflects the need for systematic benchmarking and continuous comparison of detection models [15].Recent 2026 studies further explore data-centric generalization [27], open-world detection [51], real-time communication scenarios [58], hybrid Mamba-Attention architectures [40], fine-grained tampering localization [49], and in-context learning for audio deepfake detection [11], highlighting the rapid evolution of the field.

We think a comprehensive review is necessary. This comprehensive review should trace the evolution from acoustic and deep-learning approaches, to self-supervised, transformer-based, multimodal and LLM-based methods. In particular the emergence of audio-language models presents research questions. We need to ask whether LLM-based approaches can improve generalization, interpretability, multilingual detection, robustness and adaptation to unseen deepfake generation methods. These questions are especially important because audio deepfake generation continues to outpace existing detection benchmarks.

In this review we look at research that came out from 2023 to 2026 on detecting audio and speech deepfakes. We group the studies based on how they detect audio. These methods include machine learning, deep neural networks, acoustic and spectro-temporal methods, self-supervised learning,

transformer-based models, multimodal approaches and systems using large language models or audio-language models. We also take a look at the datasets that researchers use. We examine the languages involved the ways the deepfakes are created how big the datasets are and how the studies evaluate their results. We pay attention to how well models work across different situations generalization and robustness. We also focus on detection and detection in low-resource languages. Interpretability is another point. We look at how these systems can be deployed in real-world situations. The use of language models in this field is also considered. At the end we point out weaknesses, in research and talk about where future work should go. Our goal is to help develop more reliable and more understandable audio deepfake detection systems.

2 Related Work

2.1 Conventional Deep Learning and Acoustic Feature-Based Detection

Early studies on audio deepfake detection have mostly tried to find sounds that happen when speech is created or changed. These methods usually take sound features, pictures of sound waves details about the sound waves or special sound patterns as information, for computer learning and deep learning models.

Li et al. studied voice deepfake detection using HuBERT showing that self-supervised speech representations can be powerful compared to feature-extraction methods [34]. Almutairi and Elgibreen looked at Arabic speech with self-supervised deep learning stressing how crucial it is to test detection systems on languages beyond English [5]. The NPU-ASLP system created for the ADD 2023 challenge is another example of deepfake algorithm recognition in a competition-style setting [53]. Mcuba et al. studied deep-learning techniques for investigations linking deepfake audio detection, with forensic uses [38].

Many studies have focused on spectro-temporal traits. Firc et al. Looked at spectrogram analysis to detect deepfake speech [18], At the time Bisogni et al. Studied acoustic features to make machine-learning-based audio spoofing detection more explainable [8]. Chaiwongyen et al. explored speech features as signs of synthetic speech[9]. Later they looked at features linked to timbre and used deep learning showing ongoing interest in how human perception relates to acoustic traits[10]. Falcón-López et al. also studied the selection of Mel-frequency cepstral coefficients, in detecting voice deepfakes [17].

These methods show that fake speech can have problems, with sound, frequency, timing or rhythm.. Special features and features that come from a small number of creating systems might not work well when the way speech is made changes. This leads to using learned ideas and more flexible detection systems.

2.2 Self-Supervised Learning and Pre-trained Speech Representations

Self-supervised learning has become an area of focus because pre-trained speech models can create detailed representations without needing all information to come from a limited deepfake-specific dataset. HuBERT, WavLM and other speech foundation models have therefore been studied for their ability to extract features that help tell speech apart from fake speech.

Li et al. used HuBERT to detect voice deepfake [34]. El Kheir et al. did a layer-by-layer analysis of SSL models looking at how each layer of pre-trained representations helps with audio deepfake detection [16]. Tran et al. introduced level SSL feature gating, showing how representations from different levels of a self-supervised model can be used[47]. Wang et al. looked into adapting WavLM with attention for audio deepfake detection [52]. Tahaoglu also worked on combining self-supervised learning features, with a NeXt-TDNN setup. [44].

SSL-based approaches have an advantage: they allow the detector to use speech representations that were learned from a large amount of pre-training data. However the effectiveness of these models can change depending on which layers are chosen, how fine-tuning is performed what training data is used and how similar the pre-training conditions are to the evaluation conditions. New unseen generators keep. That raises the question of whether SSL representations truly generalise or whether they mainly capture quirks that are specific, to the training distribution.

2.3 Transformer and Advanced Deep Neural Network Approaches

Transformer-based architectures are becoming more popular because they can handle long-range connections and bring together information from types of features. Gong and Li created an end-to-end transformer that uses feature fusion with cross-attention for detecting fake voices. [24]. Wani et al. developed an adaptive spectro-temporal diffusion transformer, for audio deepfake detection. This approach combines time and frequency information while focusing on being easy to understand [54]. Li et al. came up with a dual-path time-frequency attention network showing another way to mix time and frequency data[33]. Liang et al. introduced HierCon, which uses contrastive attention for detecting fake audio[36].

I notice that other architectures focus on combining representations. Cui et al. proposed fusion for robust audio deepfake detection[13]. Lee et al. Investigated reverberant waveform information through a dual-channel architecture[32]. Muruganandham et al. proposed an LSTM autoencoder architecture with residual encoding and feature fusion [39]. Wojcicki investigated learning for enhanced deepfake speech classification[39]. Wojcicki investigated ensemble learning for enhanced deepfake speech classification [55]. a movement from single-feature detectors, toward systems capable of integrating multiple acoustic or learned representations.

2.4 Multilingual and Low-Resource Audio Deepfake Detection

The ability to find deepfakes in languages is a big problem because the way people speak the sounds they make the people themselves and the data used to train systems can be different in each language. Ba and others looked at how well deepfake detection works when moving from one language to another showing that it's hard to keep the level of success when the language changes[6].

More research is now looking at languages and areas that don't have data already. Almutairi and Elgibreen worked with speech using a method that doesn't need a lot of labeled data .[5], while Ahmad et al. checked for audio in Urdu using powerful computer models [1]. Owais et al. looked at Urdu again which has little data for detecting fake audio [41]. Akhtar et al. studied speech made with special computer systems in languages like Hindi and others using their own tools[2]. Issa created a dataset called AFAD-MSA and models to find audio in Arabic [28].

These studies show that there is still a difference between languages that have a lot of data and those that do not. Even though there are multilingual data sets now the number of samples, people, ways to create the data and ways to test can still be much smaller for languages that do not have much data. Because of this being able to work with languages and being strong when moving between them are still big challenges, in research.

2.5 Datasets and Benchmarking for Audio Deepfake Detection

The quality and the variety of data sets have an impact, on what people can learn from audio deepfake detection experiments. Earlier tests had made-up speech from only a few ways to create it. As the tools to make speech got better scientists have created data sets that include ways to make speech, new types of code systems more languages and more real-life situations.

Codecfake was introduced as a dataset for finding deepfake audio that uses large language models or neural codecs. The study about Codecfake made the dataset bigger. Looked into ways to stop these deepfakes from being used[37, 57]. SpeechFake created a dataset with speech deepfakes in many languages and used the latest ways to make them [26]. SynSpeech created a dataset and a test to find speech [43]. Wang et al. made AUDETER as a dataset for finding deepfake audio when the situation is not controlled[51], while Gao et al. introduced P2V to investigate to see how well detection works when there are changes, to voices that're already in the public [21].

Benchmarking studies have also become increasingly important. Yan et al. proposed VoiceWukong as a benchmark for deepfake voice detection [59]. Dowerah et al. introduced Speech DF Arena as a leaderboard to compare speech fake detection models [15]. Firc et al. proposed an evaluation framework for comparative assessment of state-of-the-art detectors [19]. These studies show that reporting a single accuracy value in one dataset is insufficient to determine the practical effectiveness of an audio deepfake detector.

2.6 Robustness, Generalization, and Real-World Conditions

A big problem with audio deepfake detectors is that they do not work well when the sounds, around them change or when the audio is created in a way thats not known. A system that was trained on speech might not do a good job when the audio is compressed has background noise echoes goes through a phone call or is only partly changed

Several studies have therefore directly investigated robustness. Kim et al. examined inference-time noise addition for improving adversarial robustness [30]. Wang and Kim investigated denoising methods for improving the robustness of deepfake audio detection [50]. Tahaoglu incorporated multi-fused self-supervised features into a robust detection architecture [44]. Fullwood and Monrose examined how data augmentation changes the behavior of audio deepfake detectors [20]. Alali and Theodorakopoulos investigated partial fake speech attacks in real-world settings [3].

Generalization to generation systems is another big concern. Bhagtani and colleagues looked into whether recent deepfake speech generators can be detected. This work directly tackles how generation and detection are changing over time[7]. Garcia Martínez-Echevarría et al. studied how well audio deepfake detectors perform when applied to world vishing situations[22]. Huang et al. introduced a data-focused method to make speech deepfake detection work better across cases[27]. Together, these studies that robustness should be tested across generators, languages, speakers, sound environments and real-world deployment settings. It shouldn't be limited to, in-distribution test data.

2.7 Localization, Attribution, and Source Recognition

Audio deepfake detection is moving past yes-or-no answers. Instead of saying if an audio clip is fake or real researchers are now focusing on pinpointing exactly where the manipulation happened. This is called localization. They are also trying to figure out how the fake audio was made. That's known as attribution. These approaches help us understand not just if something is fake. Where and how it was created.

Wu et al.introduced a method for improving the detection and location of videos by using a step-, by-step refinement process [56]. Di Pierno et al. looked into how to find out if audio has been altered and how to recognize the model used to create it using a kind of system based on autoencoders.[14]. Wang et al.Worked on identifying the video techniques used in the ADD 2023 challenge [53]. SepVAMark explored ways to track the source of content and detect fakes by using a special kind of watermarking that combines visual and audio information[60]. These methods show that the field is needing advanced forensic tools, where simply knowing that something has been changed may not be enough.

2.8 LLM, Audio-LLM, and Multimodal Approaches

The rise of language models and audio-language models is one of the newest changes in audio deepfake detection. Old detectors usually learn a line, between real and fake audio. New LLM and audio-language-model systems give a chance to mix sound clues with words or meaning.Gu et al. presented ALLM4ADD which looks at how audio large language models can find deepfakes [25]. This study matters because it moves from deep-neural classifiers to models that can handle audio inside an audio-language framework. Chuchra and colleagues studied whether multimodal large language models can help detect deepfakes checking if combining different kinds of data helps the detection task [12].

Large language models have also been looked at as parts of systems that can adjust over time for detecting audio. Venaik and others came up with SASDN, a system that uses LLMs to keep improving at spotting speech as new types of fakes come along [48]. Of using a set-in-stone detector this method lets LLM parts help the system learn and change when new spoofing methods appear. Chou and others created ICLAD, which uses in-context learning with comparison hints to detect deepfake audio [11]. This shows another benefit of using LLM-based systems. They can learn from examples and comparisons, in context not from the usual labeled training data.

Audio-language models have also been looked into for interpretability and other tasks involving manipulation. Gohari and their team studied how to detect singing voice manipulations in a way that's easy to understand using audio-language models [23]. Thakur et al. looked at transformer-based language models for detecting deepfakes across different types of media including possible

uses, for secure media and clinical integrity [45]. Also, Kashyap et al. looked at ConLLM for finding multimodal deepfakes[29].

Even though LLM-based approaches promise much LLM-based approaches are still quite new when compared to deepfake detection methods. LLM-based approaches need a lot of computing power. LLM-based approaches rely on models. LLM-based approaches can be affected by the way a prompt is written or, by the context. LLM-based approaches may not work well with languages or generators that LLM-based approaches have not seen before. It is also necessary to see whether the reasoning skills that LLM-based approaches show actually give results than strong acoustic and SSL-based baselines. Therefore LLM-based audio deepfake detection must be judged not by how well it classifies but also by how robust it is, how easy it is to understand how well it works across different domains how much it costs to compute and how adaptable it is.

2.9 Training-Free and Foundation-Model-Based Detection

Another emerging direction is using scale pre-trained models without needing much task-specific training. Pianese et al. looked into detecting deepfake voices without any training by using these large pre-trained models [42]. This approach matters for the move toward foundation-model-based detection because it lowers the need, for big labeled datasets of deepfake audio.

The use of pre-trained speech models, SSL representations and audio-language models shows a slow shift toward foundation-model-based detection.. When comparing studies it's important to keep the difference between these methods clear. One method uses a -trained model just to get features. Another fine-tunes the foundation model. The third uses audio-language models for reasoning. These different approaches need training setups and can behave very differently when applied to new data. So the way each method is used matters a lot, for results.

2.10 Multimodal and Audio-Visual Deepfake Detection

Although this review is mostly, about audio deepfake detection some recent studies show that audio and visual information are becoming more connected. Klein and their team looked at ways to fight deepfakes using both visual data for better detection and finding exactly where the deepfake is [31]. SepVAMark studied how to combine audio data to find the source and detect deepfakes [60]. Approaches that use multimodal large language models have also been studied to find deepfakes in different types of media [12, 29].

Multimodal approaches can use types of information like speech, visual cues and meaning to improve detection. These methods can combine strengths, from each source to get results. They also need more computing power and more data to work well. With all this an audio-only detector is still important. That's because there are times when video's not available is blurry or was taken apart from the audio on purpose. So we still need tools that work with sound.

2.11 Summary of Existing Research and Research Gap

The writings from 2023 to 2026 show a change, in how people detect audio deepfakes. Early methods paid a lot of attention to features, spectrograms, regular deep learning and systems made for specific problems. Later studies started using self-supervised speech representations, transformers combining features and more advanced time and frequency designs. At the time experts have made their data sets bigger by adding speech in many languages new generation tools, neural audio codecs, situations that are not controlled and real-life noises.

Recently the field has started to focus on large language models, audio-language models in-context learning, multimodal reasoning and adaptive detection systems. Still these new approaches have not fully taken over from methods. Their benefits compared to self-supervised learning models, transformers and acoustic baselines are not clearly proven yet. There are also challenges in cross-generator generalization detecting synthesized content, across languages working with low-resource settings handling real-world signal distortions making models interpretable improving computational efficiency and keeping up with newly emerging synthesis technologies.

Therefore, an important research gap is the absence of a unifying perspective that compares the development of conventional, SSL-based, transformer-based, multimodal, and LLM/audio-language-model-based approaches while simultaneously taking into consideration the datasets, languages, generation techniques, evaluation protocols, robustness, and generalization. Addressing

this can give a more complete understanding of the current state of the art in audio deepfake detection and highlight the avenues for developing more reliable and generalizable detection systems.

3 Methodology

I carried out this literature review to find, sort and look closely at the research that exists on audio deepfake and AI-generated voice detection. I designed the method so that it could trace how detection techniques have grown from acoustic clues and basic machine learning to modern deep learning Transformer-based methods, self-supervised learning (SSL) and the newest Large Language Model (LLM) and Audio-LLM methods. While shaping the scope and categories for the studies that I looked at I also carried out an analysis of audio deepfake detection research as shown in reference [35].

The focus was, on how current methods can find AI-made voices that weren't seen before. It also looked at how these methods work when used with different data sets and languages. Another key point was how strong these methods are when used in real-life situations. These points matter because recent studies have shown that how well deepfake voice detectors work can change depending on the data set the way the voices are made and the conditions they are tested [19].

3.1 Research Questions

The review was guided by the following research questions:

- **RQ1:** Which datasets, languages and kinds of altered speech are most often used for audio deepfake detection?
- **RQ2:** what acoustic features and types of machine learning deep learning, Transformer, self-supervised LLM and Audio-LLM methods are used for audio deepfake detection?
- **RQ3:** How well existing detection methods generalize to AI-generated voices and different datasets?
- **RQ4:** How robust are current audio deepfake detection methods under real-world conditions such as noise, compression, channel distortion, and other audio processing operations?
- **RQ5:** what the major limitations and research gaps are, in audio deepfake detection research?
- **RQ6:** What potential does the combination of acoustic evidence, self-supervised speech representations, and Audio-LLM representations offer for future AI-generated voice detection systems?

3.2 Search Strategy

A systematic search was done to find studies about audio deepfake and AI-generated voice detection. The search plan focused on four ideas: audio deepfakes, synthetic speech generation, detection methods and language-model-based approaches. The search terms used were combinations of words like "audio deepfake detection " "deepfake speech detection," "AI-generated voice detection," "synthetic speech detection " "deepfake voice detection," "voice conversion detection," and "audio spoofing detection." More searches added the topic with method-related words, like "Transformer," "self-supervised learning " "large language model," "Audio-LLM " and "multimodal."

Boolean operators such as *AND* and *OR* were used to combine related keywords and improve the relevance of the retrieved studies.

3.3 Search Sources

The reviewed studies that were reviewed were gathered from academic publishers, research events, digital libraries and research collections. The final set of literature included studies linked to Elsevier, Wiley, IEEE, ACM, ACL, MDPI, Springer Nature and ICLR. These sources include areas of research connected to audio deepfake detection, such, as speech processing, artificial intelligence,

machine learning, multimedia, cybersecurity and natural language processing. Also the reference lists of studies were looked at to find more studies that are connected to audio deepfake detection and the new areas of research that are developing

3.4 Inclusion and Exclusion Criteria

The inclusion and exclusion criteria were set to pick studies that're directly relevant to audio deepfake and AI-generated voice detection. A study was included if it focused on audio, speech or voice deepfake detection and explored synthetic, generated, cloned, converted or manipulated speech. A study also qualified if it proposed, evaluated or analyzed a detection method and gave enough detail about the dataset, methodology or experimental evaluation. Additionally studies that examined analysis, deep learning, Transformer-based methods, self-supervised learning, generalization, robustness, multilingual detection, LLMs or Audio-LLMs were seen as relevant, to the review. Research articles, conference papers, journal papers and relevant preprints were included when they contained technical information. The inclusion and exclusion criteria were set to pick studies that're directly relevant to audio deepfake and AI-generated voice detection. A study was included if it focused on audio, speech or voice deepfake detection and explored synthetic, generated, cloned, converted or manipulated speech. A study also qualified if it proposed, evaluated or analyzed a detection method and gave enough detail about the dataset, methodology or experimental evaluation. Additionally studies that examined analysis, deep learning, Transformer-based methods, self-supervised learning, generalization, robustness, multilingual detection, LLMs or Audio-LLMs were seen as relevant, to the review. Research articles, conference papers, journal papers and relevant preprints were included when they contained technical information.

3.5 Data Extraction

For each selected study relevant details were. Arranged into a structured literature review matrix. The information gathered included the paper title, authors, publication year, publisher or research conference dataset used size of the dataset, language of the speech, type of speech generated Features analyzed detection method applied, model design, evaluation measures used experimental outcomes how well the model generalized robustness tested under different conditions limitations identified, gaps, in current research main contributions made by the study what was new or original and whether the dataset is publicly available. These extraction criteria were chosen because recent studies on audio deepfakes test systems using datasets, detectors and experimental setups. For example recent comparative work

[19]. Similarly, benchmark-based studies have highlighted the importance of evaluating detection models across multiple datasets rather than relying only on a single benchmark .

The information we pulled out was then used to compare the reviewed studies in an consistent way. It also helped us spot patterns see where the limits are and find gaps in the research.

3.6 Classification of Reviewed Studies

The 82 studies were sorted by their research focus and the methods they used. The main categories were:

- Feature-based and conventional machine learning methods;
- Deep learning-based methods, including CNN, RNN, LSTM and hybrid architectures;
- Transformer and self-supervised learning-based approaches
- Generalization and unseen-generator detection;
- Robustness under real-world conditions;
- Multilingual and low-resource audio deepfake detection;
- LLM, Audio-LLM, and multimodal approaches.

3.7 Study Selection

The study selection process was guided by the principles of the Preferred Reporting Items, for Systematic Reviews and Meta-Analyses also known as the PRISMA framework. This process involved four stages. The first stage was identification. The second stage was screening. The third stage was eligibility assessment. The fourth and final stage was inclusion. Each stage was carefully followed to ensure accuracy and consistency in selecting the studies. During the identification stage we gathered relevant studies by using the predefined search terms and sources. The retrieved records were then screened by looking at their titles and abstracts. Studies that were clearly irrelevant, to audio deepfake detection were excluded. The remaining studies were examined closely using the predefined eligibility criteria

After the screening and checking if they met the requirements 82 studies were chosen for the review of the literature. The whole process of selecting these studies is shown in Fig. 1.

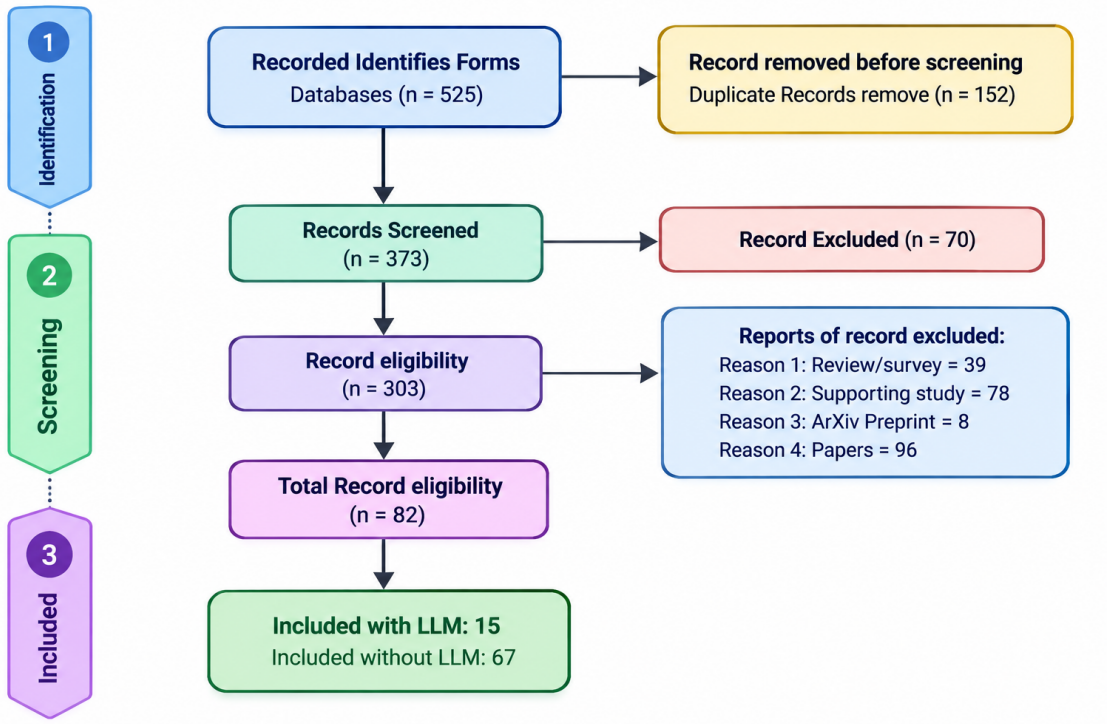


Figure 1: PRISMA flow diagram of the study selection process.

3.8 Literature Synthesis

The selected studies were looked at in a way grouping similar ideas together instead of looking at each study on its own. The analysis looked at types of data how sounds are shown in computers, how detection systems work, how they are tested how well they do, how well they can handle new situations, how strong they are and what they can't do. Special care was taken to see how well the studies worked when the data was the same as what was used to train them versus when it was different. Recent studies have shown how important it is to test deepfake detectors with new ways of making fake sounds and different sets of data [35]. Generalization to unseen voices and generation methods was therefore treated as an important dimension of the literature synthesis.

The review also examined studies focusing on real-world robustness, including noise, compression, channel effects, and other forms of audio degradation. Such evaluation is important because benchmark performance alone may not represent performance under practical conditions [19].

At the end the results were used to find the problems that still exist in research about detecting fake voices. Based on the studies that were looked at combining details from sounds using speech that learns on its own and using big language models for sound is seen as a good way to make better more flexible and easier to understand systems, for finding AI-made voices.

4 Datasets and Data Characteristics

The performance of audio deepfake detection systems is strongly influenced by the datasets used for training and evaluation. Different datasets vary in terms of language, speaker diversity, recording conditions, deepfake generation methods, and the types of attacks they contain. In recent research, benchmark datasets have been used to evaluate the ability of detection models to distinguish between authentic and synthetic speech, while newer datasets have focused on multilingual, low-resource, and real-world scenarios. Therefore, understanding the characteristics of these datasets is important for comparing existing detection approaches and identifying limitations in current research.

4.1 Major Benchmark Datasets

Several benchmark datasets have been widely used to test audio deepfake detection systems. These datasets contain speech samples and fake speech samples that are created using different text-to-speech methods, voice conversion methods and other speech generation techniques. Benchmark datasets are very helpful because they let different detection models be tested in conditions.

Table 1: Major datasets identified in the reviewed literature (2023–2026).

Dataset	Language(s)	Real/Fake	Generation/Manipulation	Main Use
ASVspoof 2019	English	Real + Fake	TTS / VC	Benchmark
ASVspoof 2021	English	Real + Fake	TTS / VC / DF	Robustness
WaveFake	Multiple	Real + Fake	TTS	Generalization
FakeAVCeleb	English	Real + Fake	TTS / VC	Multimodal
In-the-Wild	Multiple	Real + Fake	Real-world manipulation	OOD
LibriSeVoc	English	Real + Fake	VC	OOD
SONAR	English / Chinese	Real + Fake	Modern TTS	Unseen generators
Codecfake	Multiple	Real + Fake	Neural audio codec	Modern deepfake detection
SpeechFake	46 languages	Real + Fake	40 generation tools	Multilingual
AFAD-MSA	Arabic	Real + Fake	Multiple TTS	Low-resource
Urdu datasets	Urdu	Real + Fake	TTS / VC	Low-resource
BRSpeech-DF	Portuguese	Real + Fake	Multiple TTS	Low-resource
SynSpeech	Multiple	Real + Fake	Multiple synthesis methods	Benchmark
AUDETER	Multiple	Real + Fake	Multiple generation methods	Open-world
P2V	Multiple	Real + Fake	Voice perturbation	Robustness
Indic-CodecFake / SATYAM	Indic	Real + Fake	Neural audio codec	Indic languages
ADD 2023	Chinese / Multiple	Real + Fake	Multiple algorithms	Algorithm recognition
SingFake	Multiple	Real + Fake	AI-generated singing	Singing voice detection

The datasets shown in Table **??** show steps in the growth of audio deepfake detection research. Earlier benchmark datasets mainly focused on speech and speech that was made in a controlled way. Recent datasets have started to include speech in many languages, speech, from new types of generators and real-life audio. This change is important because a model that works well on one benchmark dataset might not work well on speakers, different languages, different ways of making speech or different recording situations.

4.2 Multilingual and Low-Resource Datasets

Multilingual and low-resource datasets are getting attention. I notice that many audio deepfake detection systems are mainly tested on speech. That is why recent studies have added datasets with languages such, as Arabic, Urdu, Portuguese and other low-resource languages. For example Arabic-focused datasets are used to look at detecting both speech and imitation speech. Multilingual datasets give a space to test how well systems work across languages.

The use of datasets is very important because the way sounds are made how words are said, how sounds are put together and how speech is created can be very different in different languages. Because of this a model that is trained on one language might not work well when used with another language. These datasets therefore give a chance to see if deepfake detection systems learn things, about fake speech or if they only work well for specific languages or specific data sets.

4.3 Recent and Open-World Datasets

Recent research has moved beyond test sets and is now looking at open-world and unseen-generation situations. New datasets have been made that include samples created with speech synthesis

systems and codecs so that researchers can see if detection models can spot deepfakes made by generators that were not in the training data. This direction is important because audio generation technology is developing rapidly. A detector trained with a fixed set of known generation methods may do well when the test data is similar, to what it saw but it may fail when it meets new synthesis systems. Therefore recent datasets and benchmarks are putting focus on generalization, robustness and spotting deepfakes from previously unseen generation methods.

5 Detection Methodologies

Audio deepfake detection methods have changed a lot. Audio deepfake detection methods have moved from ways of looking at sound features to using deep neural networks, self-supervised speech models, Transformer architectures and lately Large Language Models and Audio-LLMs. The studies we looked at show that each way of detecting audio deepfake detection methods looks at kinds of clues. Some look at the shape of the sound spectrum. Others look at how the sound changes over time. Some use information about the speaker. Others use speech representations that the model has learned. Some rely on the meaning of the words.. Some use evidence from more than one type of data. In this section we group the studies, by the detection method they use.

5.1 Acoustic Feature-Based Methods

Early and continuing research on audio deepfake detection has looked into characteristics that can help tell real speech apart from fake or synthetic speech. Common acoustic representations include Mel-frequency coefficients (MFCCs) spectrograms, pitch, energy, temporal characteristics, spectral information and other speech-related features.

Firc et al. looked into deepfake speech detection using spectrogram analysis showing how time-frequency representations can help find differences between speech and fake speech [18]. In the way Bisogni et al. Studied acoustic features for explainable machine learning-based audio spoofing detection [8]. Falcón-López et al. also looked into the way to choose MFCCs for voice deepfake detection pointing out that it is important to pick useful acoustic features instead of using all available coefficients without thinking [17].

Other studies have looked at specific acoustic characteristics. Chaiwongyen et al. Studied speech-pathological features, for deepfake speech detection [9]and their later work looked at pathological features connected to timbral attributes [10]. These studies show that deepfake detection can improve when using characteristics related to how speech is made and the sound quality.

Approaches that use acoustic features can be affected by recording conditions, speakers, languages and the technology used to create the speech. Handcrafted features may also not catch problems made by modern speech generation systems. So recent research is using learned representations and models that have been trained before.

5.2 Deep Learning-Based Methods

Deep learning has become a way to detect fake audio because neural networks can automatically find important features from voice signals. Convolutional neural networks, recurrent neural networks, long short-term memory networks, autoencoders and mixed designs have all been studied for this purpose.

Mcuba et al. examined at how deep learning methods work for finding audio in digital investigations [38]. Muruganandham et al. proposed an LSTM autoencoder-based parallel design that uses encoding and feature mixing for detecting fake audio. [39]. These methods show how deep neural networks can learn time-based and sound-based patterns from voice data.

Ensemble learning has also been used to mix predictions from models.. Ali et al. worked an ensemble-based approach for voice deepfake detection [4], In here,Wojcicki studied ways to classify fake speech using ensemble learning.[**?**]. Those approaches can improve robustness by combining multiple detection models or representations.

Even though these methods work well traditional deep learning models can still have differences in performance between training and real situations. Their results can drop when the generator, the person speaking the language or the place where the audio was recorded is different, from what they were trained on.

5.3 Self-Supervised and Pre-trained Speech Models

Self supervised learning has become a direction in audio deepfake detection because pretrained speech models can provide rich representations learned from large-scale speech data These representations can contain acoustic phonetic and linguistic information that may be useful for identifying manipulated speech

Li et al. investigated voice deepfake detection using the HuBERT self-supervised pre-training model [34]. Almutairi and Elgibreen similarly explored self-supervised deep learning for detecting fake Arabic speech, demonstrating the potential of pretrained speech representations in a relatively underexplored language [5].

Further research has investigated how different layers and SSL representations contribute to detection performance. El Kheir et al. presented a comprehensive layer-wise analysis of SSL models for audio deepfake detection [16]. Tran et al. proposed multi-level SSL feature gating to combine information from different representation levels [47]. Tahaoglu also investigated multi-fused self-supervised learning features with an improved NeXt-TDNN architecture [44].

Recent work has additionally explored WavLM and hybrid architectures. Wang et al. investigated adapter-based fine-tuning of WavLM using Gaussian attention [52], In other side, Ng et al. proposed XLSR-MamBo, combining Mamba and attention mechanisms for audio deepfake detection [40].

Although SSL models provide strong pretrained representations, their performance may still depend on the dataset, model layer, fine-tuning strategy and generation methods represented during evaluation.

5.4 Transformer-Based Methods

Transformer architectures have been increasingly adopted for audio deepfake detection, as their attention mechanisms can model long-range dependencies in the speech signal. In contrast to approaches based on local acoustic features, Transformer-based methods can capture long-range temporal structure.

Gong and Li proposed an end-to-end Transformer architecture combined with acoustic feature fusion through cross-attention for deepfake voice detection [24]. Wani et al. proposed an adaptive spectro-temporal diffusion Transformer that incorporates spectro-temporal information for detection [54]. Li et al. also investigated a fuzzy dual-path time-frequency attention network for audio deepfake detection [33].

Hybrid architectures have also emerged. Ng et al. investigated a Mamba-Attention backbone for scaling audio deepfake detection models [40]. These approaches demonstrate a broader movement toward architectures that combine temporal modeling, attention mechanisms, and pretrained speech representations.

However, Transformer-based models can require substantial computational resources and, despite this, still learn dataset-specific characteristics. Therefore, testing on unseen datasets and unseen generators is important in order to assess their true generalization ability.

5.5 Multimodal and Audio-Visual Methods

Deepfake detection is not about audio alone. Multimodal approaches bring together visual, linguistic or other types of information to make detection better. This is very important for videos where there might be differences, between what is being said and what is shown.

Kashyap et al. investigated multimodal deepfake detection using ConLLM, demonstrating the potential of combining multiple modalities for deepfake analysis [29]. Thakur et al. explored multimodal deepfake detection using Transformer-based large language models [45]. Klein et al. investigated audio and visual countermeasures together with fine-grained localization [31].

Multimodal systems can offer evidence that audio-only detectors cannot get.. They also need more computing power and depend on having good quality data that matches across different types of information.

5.6 LLM and Audio-LLM Based Methods

A recent development in audio deepfake detection involves the use of Large Language Models, Audio-LLMs and multimodal language models. These approaches try to go beyond binary classi-

fication by adding semantic information, contextual reasoning, comparison and interpretation.

Gu et al. worked the use of Audio Large Language Models for Audio Deepfake Detection through the ALLM4ADD framework [25]. This work shows a shift from conventional acoustic classifiers to models that can process audio representations inside a language-model-based framework.

Chuchra et al. investigated multimodal large language models for audio deepfake detection [12]. Venaik et al. proposed an LLM-integrated framework for continual adaptation in spoofed speech detection [48]. Chou et al. further explored in-context learning with comparison guidance for audio deepfake detection [11]. Gohari et al. investigated audio-language models for interpretable detection of manipulated singing voices [23].

These studies suggest that LLM-based approaches can give more than classification. They may allow reasoning over evidence comparison between speech samples, contextual interpretation and adaptation to changing conditions. However this research direction is still new. Questions remain about cost, reliability of model reasoning, dependence, on prompting or fine-tuning and whether semantic reasoning can consistently improve detection of subtle acoustic artifacts.

6 Evaluation Metrics and Benchmarking

Reliable evaluation is important when comparing audio deepfake detection systems. The studies that were reviewed use metrics to check how well classification works. These metrics include accuracy, precision, recall, F1-score Equal Error Rate and Area Under the Receiver Operating Characteristic Curve. The choice of metric and the way evaluation is done can change how detection performance is understood.

6.1 Common Evaluation Metrics

Accuracy measures how many real and fake samples are correctly classified. Even though accuracy is simple to understand it can be confusing when the data set is not balanced. Precision shows how many of the samples predicted to be fake are actually fake. Recall shows how many of the fake samples are found. The F1-score brings together precision and recall. Can give a better overall measure when the groups are not the same size. EER is often used in speech spoofing research. It is the point where the false acceptance rate and false rejection rate're the same. A smaller EER usually means performance. AUROC shows how well a detector can tell samples from fake ones at different levels of classification.

Since different studies use measures comparing models should be done with care. Firc et al. Pointed out the need, for evaluation and comparison of the best deepfake speech detectors [19]. The Speech DF Arena also demonstrates the importance of continuous benchmarking of speech deepfake detection models [15].

6.2 Cross-Dataset Evaluation

In-domain evaluation, where the training and testing data come from sources can give a too-positive view of how well a detection system works. When the data used for testing is different from the data used for training the evaluation becomes much harder. This is because the detector has to deal with data that comes from a distribution. A detector might pick up on features that're unique, to a specific dataset, recording setup, group of speakers or even the way the audio was created. For this reason testing across datasets gives a stronger sign of whether the detector can work well in real situations. It shows how well the model can adapt to unknown cases.

Li et al. emphasized the importance of understanding the relationship between generative models and detection models in audio deepfake detection [35]. Recent studies have further investigated generalization to newer generation systems and open-world conditions.

6.3 Unseen-Generator Evaluation

Modern speech generation systems evolve rapidly. Consequently, a detector trained on known generators may not necessarily detect speech generated by a previously unseen system. Bhagtani et al. investigated the detectability of recent deepfake speech generators and demonstrated the importance of considering newer generation systems [7].

Recent datasets such as AUDETER were specifically designed to investigate audio deepfake detection in open-world settings [51]. VoiceWukong also provides a benchmark for evaluating deepfake voice detection under broader conditions [59].

These studies indicate that unseen-generator evaluation should become a standard component of future benchmarking.

7 Robustness and Generalization

High performance, on benchmark datasets does not mean an audio deepfake detector will work well in situations. Real-world audio often has background noise. It can be compressed. There might be reverberation. Microphone differences can change the sound. Channels can distort the signal. Resampling can alter the audio. Other signal-processing steps can also affect the audio.. Deepfake generators keep improving. They change over time. So what works in tests may not work in the world.

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7.1 Generalization to Unseen Generators

Generalization to generation methods is one of the biggest challenges in audio deepfake detection today. A model might end up learning patterns from the generators used during training instead of picking up on general traits that show speech is synthetic.

Bhagtani et al. examined whether recent deepfake speech generators remain detectable [7]. García Martínez-Echevarría et al. further investigated the generalization gap between benchmark performance and protection against modern vishing scenarios [22]. Huang et al. proposed a data-centric approach for improving generalizable speech deepfake detection [27].

These results point to something making datasets more diverse and improving the quality of training data may matter just as much, as making the model more complex.

7.2 Noise, Compression, and Channel Effects

Real world speech is often sent through communication systems that change the sound. Compression, background noise, echoes and other things that happen in the channel can. Change the parts that a detector needs to work properly.

Kim et al. investigated inference-time noise addition for improving adversarial robustness in audio deepfake detection [30]. Wang and Kim investigated denoising methods for improving the robustness of deepfake audio detection [50]. Lee et al. considered direct and reverberant waveforms through a dual-channel detection approach [32].

These studies show that how strong a system is should be checked when there are problems, with the sound instead of only checking on clean recordings that have no issues

7.3 Perturbation and Adversarial Conditions

Deepfake detection systems can be weak when attackers make small changes to trick them. Alali and Theodorakopoulos looked at speech attacks that only change parts of the audio, in real-world situations [3]. Gao et al. introduced the Perturbed Public Voices dataset to study robust audio deepfake detection under voice perturbations [21]. This kind of work shows that future detection tools must still find deepfake signs even when the signs are not strong or have been changed.

8 Multilingual and Low-Resource Audio Deepfake Detection

Early audio deepfake detection research focused mostly on English speech. As speech synthesis tools become more common in many languages there is a growing need, for detection systems that work well in multilingual environments and in low-resource settings.

8.1 Multilingual Detection

Ba et al. into moving transferring audio deepfake detection capability across languages, demonstrating the importance of cross-language transfer [6]. Huang et al. created SpeechFake, a large-scale multilingual dataset covering 46 languages and multiple contemporary speech generation tools [26].

Multilingual datasets provide a stronger environment for determining whether a detector learns general deepfake characteristics or language-specific properties. They also make it possible to investigate whether pretrained speech representations can transfer effectively between languages.

8.2 Low-Resource Languages

Low-resource languages still get little attention in audio deepfake detection research. Almutairi and Elgibreen looked at Arabic speech by using self-supervised learning. They also created an evaluation setup that focuses specifically on Arabic [5]. Issa proposed the AFAD-MSA dataset and models for Arabic fake audio detection [28]. Ahmad et al. investigated deepfake audio detection for Urdu using deep neural networks [1], while Owais et al. further examined Urdu as a low-resource language [41].

Akhtar et al. investigated neural audio codec-generated speech deepfakes in Indic languages through Indic-CodecFake and SATYAM [2]. These studies show that the field is slowly expanding beyond English. More attention is now being paid to regions and languages. The research is starting to cover a range of geographic and linguistic areas.

8.3 Cross-Language Transfer

Cross-language transfer is still hard because different languages have sounds how they are said, the way they are spoken who is speaking and the data available to train on. A detector that is mainly trained on one language might end up learning things that're specific to that language instead of features that show if speech is fake. Systems that can detect speech in many languages should be tested in situations where they have to work across languages and with languages they have never seen before. Testing like this can show better if the detector is picking up on signs of generated speech of things that are only true, for one language.

9 Localization and Attribution of Audio Manipulations

Binary classification tells us if an audio recording is real or fake but in forensic work we need more than just that. We also need to find out which part of the audio was changed what kind of change was made and where the audio might have come from like which model or tool was used to create it. This level of detail is important, for analysis.

9.1 Temporal Forgery Localization

Wu et al. worked a coarse-to-fine framework for temporal forgery detection and localization [56]. These types approaches attempt to identify the specific temporal regions containing manipulated speech rather than assigning a single label to an entire recording.

Finding details can be very helpful when only part of a recording is fake. It can also help more when looking at the evidence, in cases where something's not real.

9.2 Source and Model Attribution

Attribution attempts to determine the source or generation model associated with a manipulated recording. Di Pierno et al. investigated audio deepfake attribution and model recognition using a multi-level autoencoder-based framework [14].

Attribution is harder, than binary detection because many generation systems can create similar artifacts. Still locating the likely source model can aid investigation and help researchers see the weaknesses of specific synthesis systems.

9.3 Watermarking and Source Tracing

Zhang et al. investigated deep separable visual-audio fusion watermarking for source tracing and deepfake detection [60]. Watermarking represents a complementary direction to passive detection because it can provide information about the origin or authenticity of generated content.

Methods that use watermarks rely on the watermark being there and working well. Passive detection is still important, for content that doesn't have a watermark.

10 Challenges and Research Gaps

The reviewed literature demonstrates significant progress in audio deepfake detection, but several important challenges remain.

10.1 Limited Generalization

A big problem is the difference between how a system works on data it has seen while being trained and how it works on data it has never seen before. Many systems show results on the test datasets that are used while they are being developed.. When they meet new situations like different people speaking, different places where the sound is recorded or different sources of data their performance often gets much worse. This means that just checking the accuracy is not enough. It does not show how dependable a system will be when it is used in life. We need methods to check how well a system works that are not about numbers, from a test set.

10.2 Dataset Diversity

Existing data sets are very different when it comes to language the variety of speakers the technology used to create them and how they were recorded. English is still better covered than many other languages. Even though new data sets, like SpeechFake, AFAD-MSA, BRSpeech-DF and Indic-CodecFake add languages, more data sets that include many languages are still needed.

10.3 Rapid Evolution of Generation Technology

Speech generation systems are changing fast. New TTS systems, voice conversion models, neural audio codecs and other generation technologies can create sounds that differ from those in datasets. This makes it hard, for detectors that rely on manipulation patterns.

10.4 Robustness in Real-World Conditions

Noise, compression, reverberation, channel distortion, and other signal-processing operations can reduce detection performance. Real-world communication platforms may also modify audio automatically. Therefore, detection systems should be tested under realistic conditions rather than only clean laboratory settings.

10.5 Interpretability

Most existing detectors focus on predicting whether an audio sample is real or fake. However, users may also need to understand why the model made a particular decision. Explainable acoustic features, localization methods, attribution techniques, and Audio-Language Models may provide opportunities to improve interpretability.

10.6 Computational Cost

Noise, compression, reverberation, channel distortion and other signal-processing operations can make it harder to detect things. Real-world communication platforms can also change audio automatically. So detection systems need to be tested in conditions instead of just in clean lab settings.

10.7 Reliability of LLM-Based Reasoning

LLM and Audio-LLM approaches show promise as a development but their reliability when used for forensic decision-making has not been studied enough. It is important to find out if the reasoning provided by language models actually offers evidence or if it only appears to work better because of specific datasets or how the prompts are designed.

11 Future Research Directions

Based on the reviewed literature, several research directions can be identified for future audio deepfake detection systems.

11.1 Generalizable Detection Across Unseen Generators

Future systems need to be built to find deepfakes that come from generation tools that were not part of the training. Open-world datasets. Checking across different generators should therefore be regular parts of how models are created. Future systems need to be built to find deepfakes that come from generation tools that were not part of the training. Open-world datasets. Checking across different generators should therefore be regular parts of how models are created.

11.2 Multilingual and Low-Resource Detection

More research is needed for languages that have limited datasets and few pretrained speech resources. Cross-language transfer, multilingual self-supervised models and zero-shot detection can help reduce the need, for language-specific datasets.

11.3 Combination of Acoustic and Semantic Evidence

Traditional acoustic features can give information about the sound spectrum, timing and rhythm of speech. Pretrained speech models and Audio-LLM models can capture abstract high-level patterns in audio. When we combine these two types of data we get a more reliable way to detect events or changes in speech. This mix of level and high-level details can make detection systems work better in different situations.

11.4 Audio-LLM and Multimodal Detection

Audio-LLMs show a way to spot audio deepfakes. Future research may explore whether Audio-LLMs can think about sound clues compare fake speech explain why Audio-LLMs flag something and keep up with the newest deepfake tools.

11.5 Real-Time Detection

Deepfake voice attacks can happen during conversations. That is why future systems need to think about detecting them and testing how well they work in real-time communication. The RTCFake benchmark is a move, in looking at how to detect fake speech during real-time calls [58].

11.6 Fine-Grained Localization and Attribution

Future detectors should go beyond labeling content as real or fake. They should be able to find the parts that have been changed figure out what kind of manipulation was used and even trace the source of how the content was made. This would make detection tools more helpful, in digital forensics and security work.

11.7 Explainable Detection

Future research should provide interpretable evidence for detection decisions. Models should ideally identify which acoustic or semantic characteristics contributed to a prediction rather than simply producing a confidence score.

This systematic literature review looked at progress in audio deepfake detection from 2023 to 2026. The studies show a shift in the field. Early methods relied on acoustic features and basic deep learning models. Over time the focus has moved toward self-supervised speech representations, Transformer-based architectures, multimodal systems and new approaches using language models (LLMs) and audio-specific large language models (Audio-LLMs). The review also shows that how well an audio deepfake detector works depends a lot on the data it is tested on. While older datasets are still used for comparison newer datasets are becoming more important. These newer datasets include speech, neural audio codecs, unknown generators, open-world scenarios and real-world audio distortions. This means researchers are starting to care about how well models work outside lab conditions. One of the challenges remains generalization. A detector that works well on data or a specific generation method might not perform as well when faced with new types of generators different languages, unknown speakers or real-world audio issues. This shows that relying on in-domain accuracy is no longer enough. Multilingual detection and low-resource language support are still areas of need. Available tools and data are focused on a few major languages. Many languages, those with small speaker populations are underrepresented. This makes it harder to build globally useful systems. Recent work on LLMs and Audio-LLMs offers a direction. These models combine based features with higher-level language understanding and reasoning. This could help detect deepfakes accurately by considering not just sound patterns but also meaning and context. However this approach is still in stages. More research is needed to check if these models are reliable interpretable, fast to run and able to work across situations. Overall the findings suggest that future audio deepfake detection systems need to be more generalizable, multilingual, robust, explainable and realistic. Combining acoustic analysis self-supervised speech models and the reasoning power of Audio-LLMs might form a strong base, for building better systems to detect AI-generated voices.

12 Conclusion

This systematic literature review examined recent developments in audio deepfake detection from 2023 to 2026. The reviewed literature demonstrates a clear progression from conventional acoustic feature-based methods and deep learning models toward self-supervised speech representations, Transformer architectures, multimodal systems, and emerging LLM- and Audio-LLM-based approaches.

The analysis also shows that the performance of audio deepfake detectors is strongly dependent on the datasets and evaluation conditions used. While established datasets remain important for benchmarking, recent datasets increasingly address multilingual speech, neural audio codecs, unseen generators, open-world conditions, and real-world perturbations. These developments highlight the need to evaluate detection systems beyond conventional in-domain accuracy.

Generalization remains one of the major challenges in the field. A detector that performs well on known datasets or generation systems may not necessarily remain effective against unseen generators, languages, speakers, or real-world audio distortions. Multilingual and low-resource detection also remains an important research area because current resources are unevenly distributed across languages.

Recent LLM and Audio-LLM studies introduce a promising new direction by combining acoustic representations with higher-level semantic and reasoning capabilities. However, this area is still developing, and further research is required to establish the reliability, interpretability, computational efficiency, and generalization of these models.

Overall, the reviewed studies suggest that future audio deepfake detection systems should move toward more generalizable, multilingual, robust, interpretable, and real-world-oriented solutions. Combining fine-grained acoustic evidence, self-supervised speech representations, and Audio-LLM-based reasoning may provide a promising foundation for developing reliable next-generation AI-generated voice detection systems.

13 Results and Findings

The systematic review of 82 selected studies shows a development in audio deepfake detection research. Earlier studies mainly focused on handcrafted features and conventional machine learning or deep learning models while recent studies increasingly use pretrained speech models, self-supervised

learning, Transformer architectures and LLM- or Audio-LLM-based approaches. The reviewed studies also show that evaluation has gradually moved from in-domain testing toward more challenging settings involving unseen generators, cross-dataset evaluation, multilingual speech and real-world audio conditions. The results indicate that acoustic feature-based methods can provide information for distinguishing real and synthetic speech particularly through features such as MFCCs, spectrograms, pitch, spectral characteristics and temporal patterns. However their performance may decrease when the recording conditions, speakers, languages or generation methods change. Deep learning models such as CNNs, RNNs and LSTMs generally provide representation learning capabilities because they can automatically learn discriminative patterns from speech data. Self-supervised and pretrained speech models have become increasingly important in studies. Models such as Wav2Vec2, HuBERT, WavLM and XLS-R provide speech representations and have demonstrated strong performance across several benchmark datasets. However the reviewed studies also indicate that high in-domain performance does not always result in cross-dataset or unseen-generator performance. Transformer-based and hybrid architectures further improve the ability to model long-range relationships and combine different types of speech representations. Recent studies also investigate Mamba-Attention and other hybrid architectures to improve scalability and representation learning. Nevertheless these models can require computational resources and may still learn dataset-specific characteristics. The review also identified an emerging group of 15 studies involving LLMs, Audio-LLMs or related multimodal language-model approaches. These studies indicate that language-model-based systems may provide capabilities such as semantic interpretation, comparison, reasoning, explanation and adaptation. However compared with deep learning and SSL-based methods, LLM- and Audio-LLM-based audio deepfake detection remains a relatively new research direction. Another important finding is the increasing focus on generalization and robustness. Several studies report that detection performance can decrease when models are evaluated on generators, different datasets, other languages or degraded audio containing noise, compression, reverberation and channel distortion. Therefore benchmark performance on a dataset may not accurately represent real-world detection capability. Overall the reviewed studies suggest that current research is gradually moving from realversus-fake classification toward more robust, multilingual generalizable, interpretable and realworld-oriented detection systems. The combination of acoustic evidence, pretrained speech representations and Audio-LLM-based reasoning appears to be a promising direction, for future research.

14 Limitations of Existing Studies

The studies we looked at show that audio deepfake detection has made progress.. There are still some problems that need to be fixed. One big problem is how much these systems depend on datasets. Many models work well when the training and testing data come from the kind of source.. When they are tested on new types of generators different datasets or real-life audio conditions their performance often drops. This means they don't always work the way outside the lab. Another issue is that the datasets used in these studies are not very diverse. Many of them focus on one language, a group of speakers a specific way of generating audio or a particular recording setup. Even though newer datasets have started to include languages and different types of generators many less common languages and unusual speech environments are still under-studied. This limits how well the models can perform in real-world situations.

Robustness is also a concern. Real-world audio often has background noise, compression, echoes changes in speed or other signal-processing effects. These changes can. Break the clues that detection models rely on. So a model that works great on controlled data might fail when used in the real world like in phone calls or online videos. Another limitation is that many models are trained on known generation techniques.. New tools like text-to-speech voice conversion, neural codecs and voice cloning are improving fast. A model trained on methods might not be able to spot newer AI-generated voices. This means detection systems need to keep up with technologies and that's not easy. Interpretability is also weak in systems. Most models just say whether an audio clip is real or fake or give a confidence number.. They don't explain why they made that decision. They don't point to acoustic patterns timing issues, spectral glitches or language inconsistencies. This lack of explanation is a problem in forensic cases or security checks, where you need proof, not just a guess. Computational cost is another issue. Big models like Transformers, multimodal systems or Audio-LLMs need a lot of memory and processing power. They also take a time to run. That makes it hard to use them in time especially on phones, tablets or other devices with limited

power. Finally models based on language models or audio-language models are still new. These models can use language reasoning to analyze audio, which is promising.. We don't know yet if they are reliable for detecting deepfakes. It's unclear whether these models can consistently find audio glitches or whether fine-tuning or prompting helps them work better across different languages, datasets or new types of generation methods. In short the research shows that current audio deepfake detection systems still have challenges. They struggle with generalization need diverse data are not robust in real settings can't handle new generation tech lack clear explanations are too costly and still need testing, for reliable use of language-based reasoning

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